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Inhibitory Roles of *Apolipoprotein E* Christchurch Astrocytes in Curbing Tau Propagation Using Human Pluripotent Stem Cell-Derived Models

Rei Murakami,[1,2] Hirotaka Watanabe,[1] Hideko Hashimoto,[1] Mayu Kashiwagi-Hakozaki,[3] Tadafumi Hashimoto,[3,4] Celeste M. Karch,[5] the Dominantly Inherited Alzheimer Network (DIAN), Takeshi Iwatsubo,[3] and Hideyuki Okano[1]

[1]Department of Physiology, Keio University School of Medicine, Tokyo 160-8582, Japan, [2]Research fellow of Japan Society of the Promotion of Science (JSPS), Tokyo 102-0083, Japan, [3]Department of Neuropathology, Graduate School of Medicine, The University of Tokyo, Tokyo 113-8655, Japan, [4]Department of Degenerative Neurological Diseases, National Institute of Neuroscience, National Center of Neurology and Psychiatry, Tokyo 187-0031, Japan, and [5]Department of Psychiatry and Hope Center for Neurological Disorders, Washington University in St. Louis, St. Louis, Missouri 63110

Genetic variants in the *apolipoprotein E* (*APOE*) gene affect the onset and progression of Alzheimer's disease (AD). The *APOE* Christchurch (*APOE* Ch) variant has been identified as the most prominent candidate for preventing the onset and progression of AD. In this study, we generated isogenic *APOE3*Ch/*3*Ch human-induced pluripotent stem cells (iPSCs) from *APOE3/3* healthy control female iPSCs and induced them into astrocytes. RNA expression analysis revealed the inherent resilience of *APOE3*Ch/*3*Ch astrocytes to induce a reactive state in response to inflammatory cytokines. Moreover, cytokine treatment changed astrocytic morphology with more complexity in *APOE3/3* astrocytes, but not in *APOE3*Ch/*3*Ch astrocytes, indicating resilience of the rare variant to a reactive state. Interestingly, we observed robust morphological alterations containing more intricate processes when cocultured with iPSC-derived cortical neurons, in which *APOE3*Ch/*3*Ch astrocytes reduced complexity compared with *APOE3/3* astrocytes. To assess the impacts of tau propagation effects, we next developed a sophisticated and sensitive assay utilizing cortical neurons derived from human iPSCs, previously generated from donors of both sexes. We showed that *APOE3*Ch/*3*Ch astrocytes effectively mitigated tau propagation within iPSC-derived neurons. This study provides important experimental evidence of the characteristic functions exhibited by *APOE3Ch/3Ch* astrocytes, thereby offering valuable insights for the advancement of novel clinical interventions in AD research.

Key words: Alzheimer's disease; *apolipoprotein E*; astrocyte; Christchurch variant; neuron

Significance Statement

Alzheimer's disease (AD) is a degenerative disease that causes cognitive decline. Familial AD is a severe form caused by mutations in *presenilin-1* (*PSEN1*), *PSEN2*, or *amyloid-beta precursor protein* genes. One carrier of the *PSEN1* mutation did not develop dementia. This carrier also had a rare variant of the *apolipoprotein E* (*APOE*) gene, the Christchurch variant. The *APOE* Christchurch variant may protect against familial AD. The mechanism of this protection is not fully understood. In the present study, we have successfully demonstrated that the *APOE* Christchurch variant suppresses the propagation of tau and exhibits a diminished capacity to convert native astrocytes into reactive astrocytes. These significant findings contribute novel insights to the field of the *APOE* gene and AD research.

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H.W.'s present address: Keio University Regenerative Medicine Research Center, Kanagawa 210-0821, Japan.

H.O.'s present address: Keio University Regenerative Medicine Research Center, Kanagawa 210-0821, Japan.

Correspondence should be addressed to Hirotaka Watanabe at hwatanabe@keio.jp or Hideyuki Okano at hidokano@keio.jp.

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Introduction

Alzheimer's disease (AD) is the most common dementia and is characterized by widespread neurodegeneration, amyloid-β (Aβ) plaques, neurofibrillary tangles, and progressive and irreversible cognitive decline (Selkoe and Hardy, 2016). Early-onset familial AD, which is caused by mutations of *presenilin-1* (*PSEN1*), *presenilin-2* (*PSEN2*), or *amyloid-beta precursor protein* (*APP*) genes in an autosomal dominant inheritance manner, has been well studied. Most AD cases are late-onset sporadic AD, and *apolipoprotein E* (*APOE*) gene *ε4* allele (*APOE4*) is the strongest genetic risk factor for sporadic AD (Alzheimer's Association, 2016). *APOE4* increases the risk of developing AD relative to the most common *APOE* variant, *APOE3*, while *APOE2* decreases this risk (Corder et al., 1993; Yamazaki et al., 2019). The *APOE* gene is expressed mainly in astrocytes in the central nervous system, and apoE proteins are secreted as lipoprotein complexes, which play an important role in lipid transportation to other neural cells. The phenotypic effects of *APOE4* and *APOE2* compared with neutral *APOE3* have been previously reported from many aspects. *APOE4* glial cells impair Aβ clearance, disrupt lipid metabolism, enhance inflammatory status, and reduce the number of neuronal spines in neurons (Lin et al., 2018; Sienski et al., 2021; de Leeuw et al., 2022; Tcw et al., 2022; Watanabe et al., 2023). In contrast, *APOE2* reduces the propensity for age-related cognitive decline, has protective effects on Aβ and tau accumulation, and is beneficial for lifespan extension independent of AD conditions (Shinohara et al., 2016, 2020).

Recent studies have reported that *APOE* rare variants may suppress AD development. *APOE3* Jacksonville variant (V236E) reduces the risk of AD and dementia with Lewy bodies by suppressing apoE aggregation and increasing lipidation of apoE lipoprotein complexes (C. C. Liu et al., 2021). *APOE* R251G variant, found in cases carrying *APOE4-R251G* alleles, attenuates *APOE4*-mediated AD pathology (Le Guen et al., 2022). Another rare *APOE* variant, *APOE3* Christchurch (Ch) variant (R136S), has recently been enthusiastically reported to suppress the progression of dementia symptoms in carriers of the *PSEN1* Colombian mutation (E280A). Although the typical *PSEN1* mutation carriers start to lose their cognitive functions in their 40s, a female individual in the pedigree did not develop dementia until she died at the age of 77, despite her apparent inheritance of the same *PSEN1* mutation. To search for unknown protective genetic factors, whole-genome sequencing analysis was conducted, and it revealed that the case carried homozygous *APOE* Ch alleles (Arboleda-Velasquez et al., 2019). Tau PET imaging showed that flortaucipir signals were confined to the occipital cortex, with comparatively less detection in the frontal cortex. Immunohistochemistry of the postmortem brain also showed that phosphorylated tau had accumulated in the occipital cortex, but to a lesser extent in the frontal cortex (Henao-Restrepo et al., 2022; Sepulveda-Falla et al., 2022). In typical cases with the *PSEN1* mutation, tau pathology accumulates throughout the brain by the age of 70. However, the tau accumulation in this case was restricted to the occipital cortex. It is unknown what prevented tau from spreading throughout the brain. The most likely candidate is *APOE* Ch variant, but there is still insufficient evidence that *APOE* Ch variant prevents tau from spreading and how *APOE* Ch variant prevents tau pathology at the molecular level.

In this study, we aimed to investigate whether *APOE* Ch variant has suppressive effects against *PSEN1* mutation-mediated AD pathology and to establish a cellular model carrying *APOE* Ch variant using human-induced pluripotent stem cells (iPSCs) that can assess tau spreading between neurons. We generated isogenic iPSCs carrying *APOE3* Ch variant from parental *APOE3* healthy control iPSCs using a CRISPR/Cas9 genome editing system and differentiated them into astrocytes. We demonstrated that *APOE3* Ch astrocytes exhibited a morphological change with a less reactive state when cocultured with iPSC-derived neurons and a suppressive effect against tau propagation among iPSC-derived neurons, mimicking the condition of the original case report. These iPSC-derived models provide novel information on the phenotypes of *APOE3* Ch astrocytes.

Materials and Methods

Culture of iPSCs

Human iPSC lines used in this study were as follows: 201B7 from healthy female donor (Takahashi et al., 2007), F12442.4 from noncarrier (PS1 H163R) healthy female donor (Karch et al., 2018), F11430.3 from *PSEN1* mutation (PS1 H163R) carrier male donor (Karch et al., 2018), and PS1-2 from *PSEN1* mutation (PS1 A246E) carrier female donor (Yagi et al., 2011). These iPSCs were cultured as feeder-free conditions with StemFit AK02N (Ajinomoto) on the iMatrix-511 (Laminin-511 E8, Nippi)-coated culture plates in an atmosphere containing 3% CO_2. The feeder-free iPSCs were passaged once a week as the following procedure: iPSCs were detached by 0.5× TrypLE Select (Thermo Fisher Scientific) diluted with phosphate-buffered saline (PBS, Nacalai) and seeded with the density of 1.3×10^4 cells/well in 6-well culture plates with AK02N in the presence of 10 μM Y-27632 (Nacalai). The culture media were changed the next day from the passage and every 2 d.

For the iPSCs in on-feeder conditions, iPSCs were cultured on mitomycin C-treated SNL murine fibroblast feeder cells with the hES medium and Dulbecco's Modified Eagle Medium/Ham's F-12 with phenol red, HEPES, and sodium pyruvate (DMEM/F12, Fujifilm) including 20% KnockOut serum replacement (KSR, Life Technologies), 2 mM L-glutamine (Invitrogen), 0.1 mM nonessential amino acids (NEAAs, Nacalai), 0.1 mM 2-mercaptoethanol (Sigma-Aldrich), 4 ng/ml fibroblast growth factor 2 (PeproTech), and penicillin/streptomycin in an atmosphere containing 3% CO_2. The culture media were changed every day. The on-feeder iPSCs were passaged once a week with the following procedure: iPSCs were detached by dissociative solution [0.25% trypsin, 100 μg/ml collagenase IV (Invitrogen), 1 mM $CaCl_2$, and 20% KSR] and seeded on the feeder cells with the hES medium.

Generation of isogenic iPSC line

Construction of the APOE3*Ch targeting vector and single guide RNA (sgRNA) plasmid vectors.* The targeting vector for *APOE ε4* gene, which contains a puromycin *N*-acetyltransferase gene (Pac), thymidine kinase (TK), and two TTAA tetranucleotides encompassing the Pac/TK (TTAA-Pac-TK-TTAA), was used as a template (Watanabe et al., 2023). Thymine at *rs429358* and cytosine at *rs121918393* within exon 4 in the *APOE ε4* targeting vector were converted to cytosine and adenine, respectively, by KOD -Plus- Mutagenesis Kit (TOYOBO) with two primer pairs: 5′-CACACGTCCTCCATGTCCGCGCCC-3′ and 5′-GCACACGTCCTCCATGTCCGCGCC-3′ for converting from thymine to cytosine at *rs429358* and 5′-GAGCCTCGCCTCCCACCTGCG CAAG-3′ and 5′-ACCCGCAGCTCCTCGGTGCTCTG-3′ for converting from cytosine to adenine at *rs121918393*.

Two sgRNA plasmids (Watanabe et al., 2023) expressing both sgRNA and *S. pyogenes* Cas9 nuclease (SpCas9) were used for the genome editing.

Generation of the isogenic APOE3*Ch/3Ch iPSCs.* For the generation of the isogenic iPSCs carrying the targeted *APOE3*Ch/3Ch alleles, 201B7 line iPSCs carrying *APOE3/3* were electroporated to make the *APOE3*Ch targeting vector plasmid and two sgRNA plasmid vectors pass through the iPSC cell membranes. During the electroporation, iPSCs were suspended with Opti-MEM (Invitrogen) containing 10 μM Y-27632 and

cultured in AK02N containing 20 μM Y-27632 for 1 d. The media were changed with fresh AK02N containing 10 μM Y-27632 next day after electroporation, and the iPSCs were treated with 0.7 μg/ml puromycin in AK02N 2 d after electroporation for selecting iPSCs carrying homozygous knock-in alleles. Three days after electroporation, the culture media were changed with fresh AK02N without puromycin and changed with AK02N containing 1.0 μg/ml puromycin 5 d after electroporation. Ten days after electroporation, the media were changed with AK02N without puromycin and changed with a new one every other day.

Thirteen days or more after electroporation, surviving iPSC colonies were picked up and cultured separately by clones. Genomic DNAs were extracted from the picked iPSCs and amplified by polymerase chain reaction (PCR) using the following primer pair: 5′-GGGCTGCGTTGC TGGTCACATT-3′ and 5′-ACCCCATCTCCACACACACCAC-3′. The amplified PCR products were used as the next nested PCR templates. The primary PCR products were amplified by nested PCR using the following primer pair: 5′-AAGAAGCATTTGTGGAGCACC-3′ and 5′-TGAATCTTTATTAAACTAGG-3′. Some clones having homozygous 4,795 bp of nested PCR products, which contained TTAA-Pac-TK-TTAA between exons 3 and 4 at agarose gel electrophoresis, were selected. Forty-six iPSC clones carried the Pac-TK gene cassette out of 84 picked iPSC clones. To confirm whether *APOE3*/*3* was replaced by *APOE3*Ch/*3*Ch in the clones, the selected iPSC clones were verified exon 4 by sequencing with the following primers: 5′-TCTCTGCATCTGTCTCTGTC-3′, 5′-TGAATCTTTATTAAACTA GG-3′, and 5′-TTCACCTCGTCCAGGCGGTC-3′. Finally, one clone was revealed to carry the homozygous *APOE3*Ch/*3*Ch variant correctly.

The clone identified as carrying the homozygous *APOE3*Ch variant was treated with piggyBac transposase expression vectors (Watanabe et al., 2023) to remove the Pac-TK gene cassette. PiggyBac transposase expression vectors were transfected into the selected iPSC clones with transIT-LT1 reagent (MiRus) and treated with 10 μM ganciclovir 2 d after transfection. Seven days after transfection, the media were changed to the fresh one without ganciclovir, and 13 d or more after transfection, surviving iPSC clones were picked up. Genomic DNAs from the picked clones were amplified by primary and nested PCR as described above. Two iPSC clones out of eight picked clones were identified as completely lacking Pac-TK gene cassette by agarose gel electrophoresis and sequencing with a primer: 5′-TCCATCAGCGCCCTGCGGCC-3′.

The karyotype analysis for isogenic iPSCs. G-banding analysis for the karyotype check of the isogenic iPSCs was performed by LSI Medience. The karyotype of the parental *APOE3*/*3* iPSC clone (201B7) is normal with 46XX (Takahashi et al., 2007).

Differentiation from iPSCs to astrocytes

Astrocytes were induced from iPSCs by the iPSC-derived astrocyte induction protocol which we previously reported (Leventoux et al., 2020). On-feeder iPSCs were detached to form embryoid bodies (EBs). EBs were dissociated into single cells to form neurospheres (NSs). NSs were also dissociated into single cells and attached on culture dishes coated with Matrigel (Corning). Attached cells were cultured with MHM containing B27 (Invitrogen), NEAA, brain-derived neurotrophic factor (BDNF, R&D), and glial cell line-derived neurotrophic factor (GDNF, Alamone) and passaged at least three times before experiments. Media for iPSC-derived astrocytes were changed once a week.

Differentiation from iPSCs to neurons

Forebrain cortical neurons were induced from iPSCs by the robust neuronal induction protocol that we previously reported (Sato et al., 2021; Watanabe et al., 2021). iPSCs were treated with SMAD inhibitors [SB431542 (Tocris) and LDN193189 (StemRD)] and Wnt inhibitor (IWP-2, Sigma) to form neural progenitor cells (NPCs). NPCs were differentiated into neurons with neural medium-containing DAPT (Sigma-Aldrich) for 5 d, and then the culture media were changed to the BrainPhys neuronal medium (STEMCELL Technologies) containing BDNF, GDNF, ascorbic acid (Sigma-Aldrich), dbcAMP (Sigma), and PD0332991 (Sigma-Aldrich). Half volume of media for iPSC-derived neurons were changed every 3–4 d.

Immunocytochemistry

iPSC-derived astrocytes and iPSC-derived neurons were fixed with 4% paraformaldehyde (PFA) and blocked with PBS containing 5% fetal bovine serum (Invitrogen), and 0.1% Triton X-100 (Nacalai) for 2 h at room temperature and incubated with primary antibodies overnight at 4°C. The primary antibodies were listed as follows: rat anti-GFAP (1:500, Invitrogen, catalog #2.2B10), rabbit anti-EAAT1 (1:200, Abcam, catalog #ab416), mouse anti-S100β (1:500, Sigma-Aldrich, catalog #S2532), mouse anti-FLAG (1:500, Sigma-Aldrich, catalog #F1804), mouse anti-βIII tubulin (1:500, Sigma-Aldrich, catalog #T8660), rabbit anti-tau (DAKO, 1:1,000, Agilent, catalog #A0024), and rabbit anti-LRP1 (1:1,000, Abcam, catalog #ab92544). Cells were washed with PBS and incubated with appropriate Alexa Flour (488, 555, or 647)-conjugated secondary antibodies (Invitrogen) and Hoechst 33258 solution (Wako) for 1 h at room temperature. Fluorescent signals were detected by a confocal microscope (LSM700, ZEISS) and IN Cell Analyzer 6000 (Cytiva).

Calcein staining

iPSC-derived astrocytes and iPSC-derived neurons were incubated with the media containing 1 μg/ml calcein-AM (DOJINDO) at 37°C for 1 h before immunocytochemistry.

L-glutamate uptake assay

L-glutamate concentrations were measured by L-glutamate Assay Kit YAMASA NEO (YAMASA). iPSC-derived astrocytes were plated with the 1.0×10^5 cells/well in 96-well culture plates, and iPSCs were plated with the 1.0×10^4 cells/well in 96-well culture plates and cultured for 1 week before the assay. For the L-glutamate assay, the media were changed to AK02N containing 1 mM L-glutamate and incubated for 4 h. Media containing L-glutamate were sampled at 0, 1, 2, and 4 h after incubation started, and the L-glutamate concentration in the sampled media was measured by following the L-glutamate Assay Kit YAMASA NEO procedures.

Cytokine treatment to iPSC-derived astrocytes

To induce iPSC-derived astrocytes to the reactive state, cytokine cocktail [30 ng/TNFα (PeproTech), 3 ng/ml IL-1α (PeproTech), and 400 ng/ml C1q (Prospec)] was treated to iPSC-derived astrocytes alone or cocultured astrocytes for one overnight incubation at 37°C.

Quantitative reverse transcription-PCR (qRT-PCR)

The total RNA was purified by RNeasy Mini Kit (QIAGEN) and reverse-transcribed with random hexamers. qPCR was conducted using TB Green Premix Ex Taq II (Takara Bio) by ViiA7 system (Applied Biosystems) with cDNA and gene-specific primers. Threshold cycle values of the interest were normalized to those of *GAPDH*. The primer pairs used in this study were shown as follows: 5′-TGGCCCTCATCGACG TTTTC-3′ and 5′-GCAGGTAATCCCAAAAGCGAC-3′ for *APOE*, 5′-GGGGAGTCCCATGTACTCTATC-3′ and 5′-GGAAGTCGTGG ACAGTAACAG-3′ for *C3*, 5′-GTGGCATTCAAGGAGTACCTC-3′ and 5′-TGATGGCCTTCGATTCTGGATT-3′ for *CXCL10*, 5′-CTATC TGCAATTACGCAGCCT-3′ and 5′-TGTTCTGGCTTCTTGGGATG A-3′ for *GBP2*, 5′-CATCTCCTTGCGATTCTGTTTTG-3′ and 5′-CC ATTCCGAGTGCTCCTGA-3′ for *PTX3*, 5′-AGGCTGAGAACGGG AAGCTT-3′ and 5′-ACTCCACGACGTACTCAGCG-3′ for *GAPDH*.

Lipid staining

iPSC-derived astrocytes were treated with oleic acid (OA) before lipid staining to confirm whether a lipid staining reagent works. OA was dissolved in dimethyl sulfoxide (DMSO, Nacalai) with 500 mM concentration. A 500 mM OA was diluted for 4 mM with 0.14 g/ml bovine serum albumin (BSA, Nacalai), which was dissolved in Tris–HCl buffer (Nacalai), pH 8.0, as 0.14 g/ml concentration and filtered with 0.22 μm pore. A 4 mM OA or DMSO in BSA–Tris buffer were diluted with DMEM/F12 (final 200 μM OA). iPSC-derived astrocytes were incubated with 200 μM OA or DMSO for one overnight, and after incubation, the cells were treated with LipiBlue (final 100 nM, Wako), calcein-AM (final 1 μg/ml, Wako), and DRAQ5 (final 5 μM, BioStatus) in DMEM/F12 for 1 h at 37°C incubation. After the incubation with staining reagents, the

iPSC-derived astrocytes washed with DMEM/F12 and fixed with 4% PFA and fluorescent signals were detected by IN Cell Analyzer 6000. The stained lipid particles in the iPSC-derived astrocytes were counted by IN Cell Analyzer 6000.

Enzyme-linked immunosorbent assay (ELISA) for apoE

Media for iPSC-derived astrocytes were collected and centrifuged to remove floating cells and insoluble substances. The remained cells were lysed by RIPA buffer containing cOmplete protein inhibitor cocktail (Sigma-Aldrich) and dithiothreitol (DTT). Total protein amount of these cell lysates was evaluated by the BCA Protein assay (Thermo Fisher Scientific).

ApoE in the conditioned media were quantified by ELISA with the following procedures; a 96-well culture plate was coated with mouse anti-apoE antibodies (Novus Biologicals, catalog #NB110-60531) in a 100 mM carbonate buffer at 4°C for one overnight. The antibody-coated plates were blocked with 1% nonfat milk in PBS and washed with TBST buffer (Nacalai). Recombinant apoE3 protein (PeproTech) was used as a standard quantification for ELISA. The conditional media were diluted and applied to the antibody-coated plate and incubated one overnight at 4°C. The plate was washed and incubated with biotin-conjugated goat anti-apoE antibodies (Meridian Life Science, catalog #K74180B) for 2 h at room temperature. The plate subsequently was treated with streptavidin–horseradish peroxidase (Fitzgerald Industries) for 90 min at room temperature, and tetramethylbenzidine liquid substrate was added for ELISA Super Slow (Sigma-Aldrich) for 5 min incubation at room temperature. The enzyme–substrate reaction was stopped by 1 M H_2SO_4, and the optical density at 450 nm of the plate was measured by the EnSpire multimode plate reader (PerkinElmer). Each apoE concentration was normalized by each total protein expression level measured by the BCA Protein assay (Pierce).

ELISA for total tau, phosphorylated T181 tau, and phosphorylated T231 tau

Differentiated iPSC-derived neurons were cultured for 2 weeks, followed by coculture with or without iPSC-derived astrocytes for 2 d. The cultures were incubated in media containing potassium chloride (KCl) buffer (in mM: 55 KCl, 10 HEPES–HCl, 5.2 NaCl, 0.4 $MgCl_2$, 6.9 glucose, 0.9 $CaCl_2$) or reference buffer (in mM: 2.5 KCl, 10 HEPES-HCl, 55 NaCl, 0.4 $MgCl_2$, 6.9 glucose, 0.9 $CaCl_2$) for 1 h at 37°C (Pooler et al., 2013). Conditioned media were collected and centrifuged to remove floating cells and insoluble substances, and cells on the culture plates were lysed by RIPA buffer containing cOmplete protein inhibitor cocktail and DTT for the BCA Protein assay.

The levels of total tau in the conditioned media were quantified by the Human Tau (Total) ELISA Kit (Invitrogen) with the following kit protocol; the conditioned media were applied to the anti-human total tau antibody-coated culture plates and incubated for 2 h at room temperature. After incubation, culture wells were washed, and the biotin conjugated anti-human total tau antibodies were applied to the wells with 1 h incubation at room temperature. Then wells were washed, and streptavidin–HRP were added to the wells for 30 min at room temperature. The wells were washed and treated with stabilized chromogen solution for 30 min incubation at room temperature in the dark. Stop solutions were added to the wells filled with the stabilized chromogen solution after the incubation. The optical density at 450 nm of the wells was measured by the EnSpire multimode plate reader, and the total tau concentration of each well was calculated with OD_{450} and normalized by each total protein expression level measured by the BCA Protein assay.

The levels of phosphorylated tau in the conditioned media were quantified by Phosphorylated Tau T181 ELISA Kit Wako (Wako) or Tau (Phospho) [pT231] Human ELISA Kit (Invitrogen), according to the manufacturer's instructions. After completion of the reaction, luminescence (T181) or optical density at 450 nm (T231) was measured by the EnSpire multimode plate reader. The levels of phosphorylated tau concentrations of the wells were calculated and normalized by each total protein level measured by the BCA Protein assay.

Immunoprecipitation for apoE lipoprotein complexes

For preparation of immunoprecipitation, streptavidin-conjugated agarose beads (Sigma-Aldrich) were incubated with biotin-conjugated goat anti-apoE antibodies (Meridian Life Science, catalog #K74180B) in advance for 2 h at room temperature with a rotator.

The anti-apoE antibody-linked agarose beads were mixed with the conditioned media of the iPSC-derived astrocytes one overnight at 4°C with mixing by a rotator. The mixed beads were centrifugated to separate beads from the media and the media removed. The beads were washed with TBS buffer and centrifugated to remove the buffer. The beads were treated with TBS containing 0.1% Triton X-100 (Nacalai, TBST buffer), and the lipid components of apoE lipoprotein complexes were eluted into TBST buffer as a lipid fraction. Subsequently, the beads were treated with 100 mM glycine–HCl, pH 2.5, and apoE protein was eluted as an apoE fraction. The apoE fraction was immediately added with 1 M Tris–HCl, pH 8.8, to neutralize the fraction.

Cholesterol assay for apoE lipoprotein complexes

The amount of cholesterol in lipid fraction by immunoprecipitation was measured by the Amplex Red Cholesterol Assay Kit (Invitrogen). Experiments to measure cholesterol levels were conducted following the procedures of the Amplex Red Cholesterol Assay Kit (Thermo Fisher Scientific). The fluorescent signals ($\lambda_{excitation}$, 550 nm; $\lambda_{emission}$, 590 nm) were measured and quantified by the EnSpire multimode plate reader. The cholesterol concentrations were normalized by apoE levels of apoE fractions measured by apoE ELISA.

Morphology assay for iPSC-derived astrocytes

iPSC-derived astrocytes were stained by immunocytochemistry with anti-GFAP antibody, and the fluorescent signals were detected by a confocal microscope with the procedures described above. Images of GFAP-stained astrocytes were processed in the Fiji ImageJ software. Background signals in the astrocytic images were erased with ImageJ eraser tools, and the remaining astrocyte shapes were analyzed by Sholl analysis by the following procedure: counting intersections between concentric circles and astrocytic images. The concentric circles were drawn every 5 μm from the centers of the astrocytic soma.

Human tau propagation assay

Two-week cultured iPSC-derived neurons (1.5×10^5 cells/well in 48-well culture plates) were transfected with human tau vectors, which were accompanied by human synapsin I promoter and a tandem sequence of nuclear localization signal (NLS)-dTomato, P2A, and 3× FLAG-human tau repeat domain (RD, residues 244th–372nd), by the Lipofectamine™ 2000 transfection reagent (Invitrogen). Two days after transfection, dTomato expression was detected with a fluorescent microscope. Then iPSC-derived astrocytes passaged onto the transfected iPSC-derived neurons (1.3×10^4 cells/well). Four days after transfection, KCl treatments to iPSC-derived neurons were conducted for 1 h to depolarize neuron membrane potential and increase tau secretion from neurons (Pooler et al., 2013).

After the KCl treatments, the iPSC-derived neurons were cultured with the media for iPSC-derived neurons for 1 d. iPSC-derived neurons were washed with PBS and fixed with 4% PFA for one overnight at 4°C. FLAG and βIII tubulin in the fixed iPSC-derived neurons were stained by immunocytochemistry. FLAG- and dTomato-positive puncta in the vector-transfected iPSC-derived neurons, and dTomato-negative FLAG-positive puncta, which indicate untransfected but tau-propagated iPSC-derived neurons, were counted, and the ratio of tau propagation was calculated with the numbers of these puncta. For the validation of transfection efficacy, dTomato- and Hoechst-positive puncta per Hoechst-positive puncta were counted, and the transfection efficacy was calculated.

During the tau propagation assay, 30 nM receptor-associated protein (RAP, Oxford Biomedical Research) was treated to the vector-transfected neurons at the same period of KCl treatment. After the treatments, the neurons were fixed and stained with immunocytochemistry as described above. The ratio of tau propagation was also calculated.

Experimental design and statistical analyses

For all cell preparations and experiments, samples were prepared with independent three or more culture batches. iPSC lines used in this study were the following: two isogenic lines (clones #1 and #7) carrying the *APOE3*Ch/*3*Ch, *APOE3*/*3* one parental healthy control line (201B7), another healthy control line (PS1 H163R noncarrier), and two *PSEN1* mutant lines carrying PS1 H163R or PS1 A246E mutation.

Data are presented as the mean value ± standard error of the mean (SE) unless mentioned. The data in Figures 2*J*, 4*E*, 5*B–F*, and 6*B,C* were analyzed by one-way analysis of variance (ANOVA) and post hoc Tukey's test. The data in Figures 2*E* and 3*A–C,F–H* were analyzed by two-way ANOVA. The data in Figures 2*D,F–H* and 4*B* were analyzed by unpaired *t* test. Statistical significance was defined as $p < 0.05$. The data were analyzed by the Prism software (GraphPad Software, Ver.10.0.0).

Results

The generation of isogenic iPSCs carrying *APOE3*Ch/*3*Ch by CRISPR/Cas9 genome editing system

We aimed to analyze the *APOE3*Ch/*3*Ch cellular and molecular functions in vitro to understand the resilience of *APOE3*Ch variant to AD development and seek a new therapy for clinical AD patients. We first generated isogenic *APOE3*Ch/*3*Ch iPSCs from healthy control *APOE3*/*3* iPSCs (201B7 line) with CRISPR/Cas9 genome editing system to match the genetic background except for the specific genetic site. The *APOE3*Ch targeting vector and two SpCas9 nuclease–sgRNA vectors were electroporated into *APOE3*/*3* control iPSCs, followed by puromycin treatments for colony selection. The genomic DNAs extracted from selected iPSC clones were amplified by nested PCRs and verified by agarose gel electrophoresis whether the iPSC genomes included the targeting vector sequence (Fig. 1*A,B*). Targeted iPSC clones were confirmed as being converted from *APOE3*/*3* to *APOE3*Ch/*3*Ch by direct sequencing (Fig. 1*C*). Next, the Pac–TK selection gene cassettes in targeted iPSC clones were removed by a piggyBac transposase, and the iPSCs were selected with ganciclovir for the exclusion of iPSCs carrying the selection gene cassette (Fig. 1*A*). The resultant eight iPSC clones were picked up, and the genomic DNAs extracted from the picked iPSC clones were checked by nested PCRs whether the selection gene cassettes were excised (Fig. 1*D,E*). Two iPSC clones (#1 and #7), which carried the homozygous *APOE3*Ch/*3*Ch and lacked the Pac/TK selection cassette correctly, were analyzed for the karyotypes with G-banding analysis and confirmed to have normal karyotypes (Fig. 1*F*). We concluded that the generation of isogenic *APOE3*Ch/*3*Ch iPSCs is successful. The two isogenic *APOE3*Ch/*3*Ch clones presented the same tendencies in the following explements, and the results of the two clones were combined to distinguish the two clones; the clones were labeled in gray (clone #1) or white (clone #7) in the results of the subsequent analysis.

Differentiation to astrocytes carrying *APOE3*/*3* or *APOE3*Ch/*3*Ch from parental and isogenic iPSCs and characterization of apoE and lipid in *APOE3*/*3* and *APOE3*Ch/*3*Ch iPSC-derived astrocytes

In the brain, apoE lipoprotein complexes are secreted mainly from astrocytes. We differentiated a parental *APOE3*/*3* iPSC and two isogenic *APOE3*Ch/*3*Ch iPSCs to astrocytes by the induction protocol we previously reported (Leventoux et al., 2020; Fig. 2*A*). The differentiated cells formed astrocytic morphologies (Fig. 2*B*) and expressed astrocytic markers such as GFAP, EAAT1, and S100β (Fig. 2*C,D*). We next measured the time course change of L-glutamate concentration in the conditioned media of the differentiated cells and iPSCs to assess one of the astrocytic physiological functions, the uptake of excitatory neurotransmitter L-glutamate via the glutamate transporter. The L-glutamate concentration in *APOE3*/*3*- and *APOE3*Ch/*3*Ch-differentiated culture media decreased in a time-dependent manner until 4 h of incubation, whereas there was no change in iPSC culture media (Fig. 2*E*). These data suggested that the differentiated cells have astrocytic functions, and we can use these cells as astrocyte models in subsequent experiments.

We evaluated the expression and secretion of apoE lipoprotein complexes. The result of qRT-PCR showed that *APOE* expression was the same level between *APOE3*/*3* and *APOE3*Ch/*3*Ch astrocytes (Fig. 2*F*). ELISA for apoE in the conditioned media revealed that *APOE3*/*3* and *APOE3*Ch/*3*Ch astrocytes had no difference in secretion of apoE protein (Fig. 2*G*), and cholesterol assay also clarified that both genotype astrocytes had no difference in secreted apoE-associated cholesterol levels (Fig. 2*H*). ApoE lipoprotein complexes secreted from astrocytes contribute to lipid transportation and homeostasis in brain environments. We evaluated the functions of *APOE3*Ch/*3*Ch iPSC-derived astrocytes associated with lipid metabolisms. To evaluate lipid generation and storage in the astrocytes, we stained lipid in the astrocytes with lipid-staining reagents. Lipid in astrocytes formed particles like lipid droplets, but not like fat reservoir in fat cells (Fig. 2*I*). We counted the number of lipid particles and found that the same number of lipid particles was formed in *APOE3*/*3* and *APOE3*Ch/*3*Ch astrocytes. OA treatment increased the number of lipid particles in the astrocytes but did not make significant differences between *APOE3*/*3* and *APOE3*Ch/*3*Ch astrocytes (Fig. 2*J*). These results showed that *APOE3*Ch/*3*Ch astrocytes have no change of intracellular lipid accumulation and apoE lipoprotein secretion (Fig. 2*G–J*).

Molecular and cellular alterations in *APOE3*/*3* and *APOE3*Ch/*3*Ch astrocytes

To validate the features of *APOE3*/*3* and *APOE3*Ch/*3*Ch astrocytes with an aspect to reactive states, we performed qRT-PCR for several genes associated with astrocytic reactive states (Fig. 3*A–C*). iPSC-derived astrocytes were normally maintained with media without microglial cytokines, which can stimulate astrocytes to induce reactive astrocytes (Liddelow et al., 2017; Hasel et al., 2021). We treated recombinant microglial cytokines (TNFα, IL-1α, and C1q) to *APOE3*/*3* and *APOE3*Ch/*3*Ch astrocytes and validated gene expressions associated with reactive astrocytes.

The expression levels of pan-reactive astrocyte marker genes in the *APOE3*/*3* astrocytes were significantly enhanced by the cytokine treatment, while the expression levels of the genes in the *APOE3*Ch/*3*Ch astrocytes were not enhanced significantly by the cytokine treatment (Fig. 3*A*). Next, the expression levels of inflammatory state-specific (A1-specific) and neurotrophic state-specific (A2-specific) genes in *APOE3*/*3* and *APOE3*Ch/*3*Ch astrocytes were also validated. The cytokine treatment enhanced these gene expressions in only *APOE3*/*3* astrocytes, but *APOE3*Ch/*3*Ch astrocytes did not change the expressions of these genes significantly with the treatment (Fig. 3*B,C*). These results suggest that *APOE3*Ch/*3*Ch astrocytes have a resistance to elicit inflammatory or neurotrophic reactive astrocytes in response to the cytokine treatment.

To evaluate microglial cytokine-induced changes in the astrocytes and interactions among astrocytes and neurons, we developed a coculture system between iPSC-derived neurons and astrocytes with or without cytokine treatment.

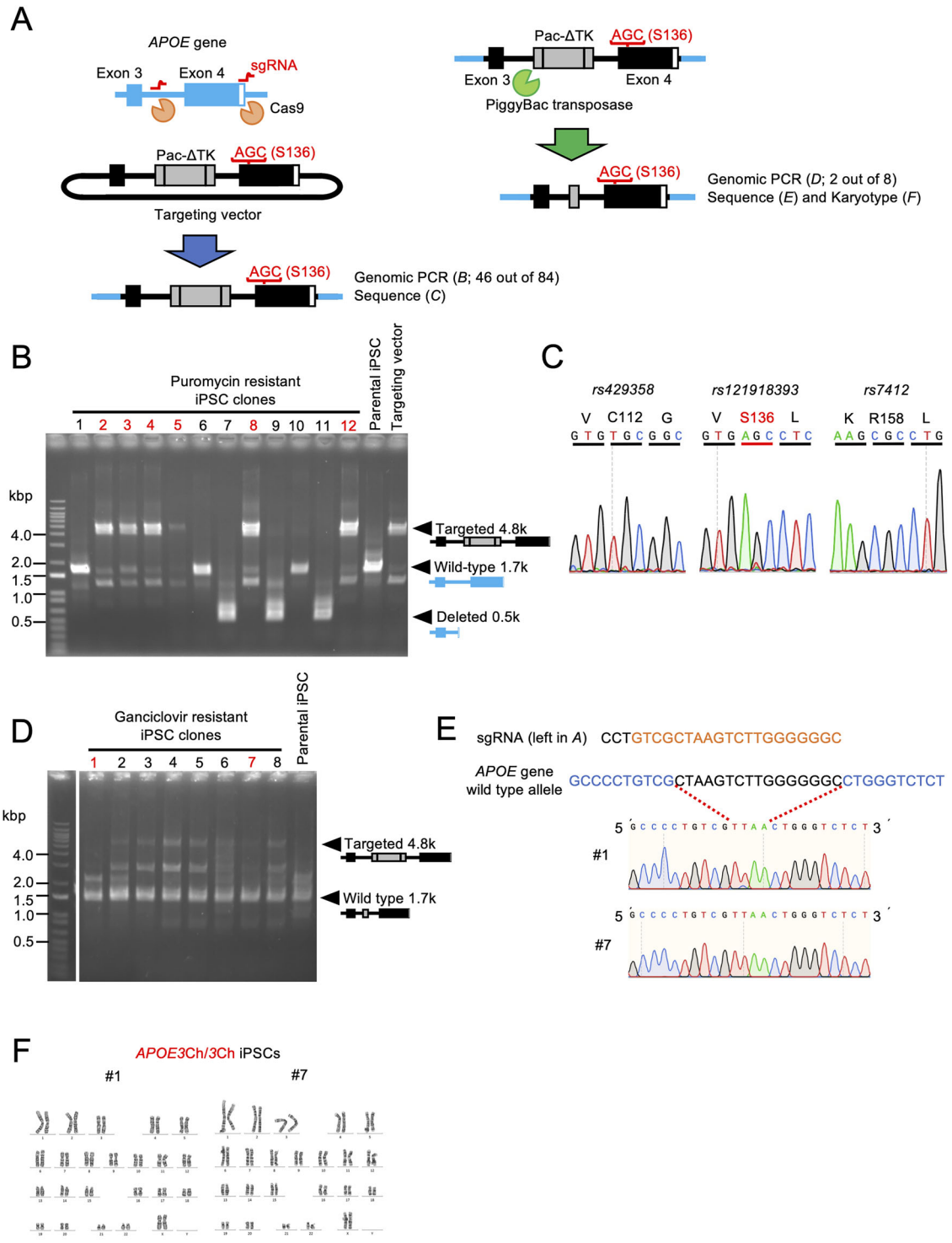


Figure 1. Generation of isogenic *APOE3*Ch/*3*Ch iPSCs from parental *APOE3/3* iPSCs with CRISPR/Cas9 genome editing system. ***A***, Schematic experimental paradigm for genome editing using CRISPR/Cas9 system. The targeting vector includes *APOE* gene exons 3 and 4 (black boxes) and the PGK Pac-ΔTK cassette (long gray box) sandwiched between two *ITR* sequences (short gray boxes) for removal by piggyBac transposase. Cyan boxes and lines indicate the iPSC genomes. Puromycin treatment selects iPSC clones successfully carrying knock-in alleles including puromycin-resistant genes. After genomic PCR and sequencing to confirm that selected iPSC clones carry *APOE* Ch variant, the confirmed clones are treated with piggyBac transposase to remove the PGK Pac-ΔTK cassette, and then ganciclovir treatment selects iPSC clones with *APOE* Ch variant and without the PGK Pac-ΔTK cassette. The screening method and the number of iPSC clones were also delineated within diagrams, accompanied by pertinent subpanel label. ***B***, A representative image of genomic PCR for *APOE* alleles after puromycin treatment is shown. Nested PCR uncovers that iPSC clones indicated with red numbers carry homozygous targeted alleles. ***C***, *APOE* allele of Number 5 clone is analyzed by sequencing in ***B*** around the polymorphism *rs429358*, *rs121918393*, and *rs7412*. ***D***, A representative image of genomic PCR for *APOE* alleles after ganciclovir treatment to Number 5 clone in ***B***. Nested PCR unveils iPSC clones indicated with red numbers without the PGK Pac-ΔTK cassette. ***E***, Clone numbers #1 and #7 show correct excision of the selection cassette in the intron 3 of the *APOE* gene. ***F***, Karyotypes of clone numbers #1 and #7 in ***D*** are analyzed by the G-band staining method. Both clones show normal karyotypes.

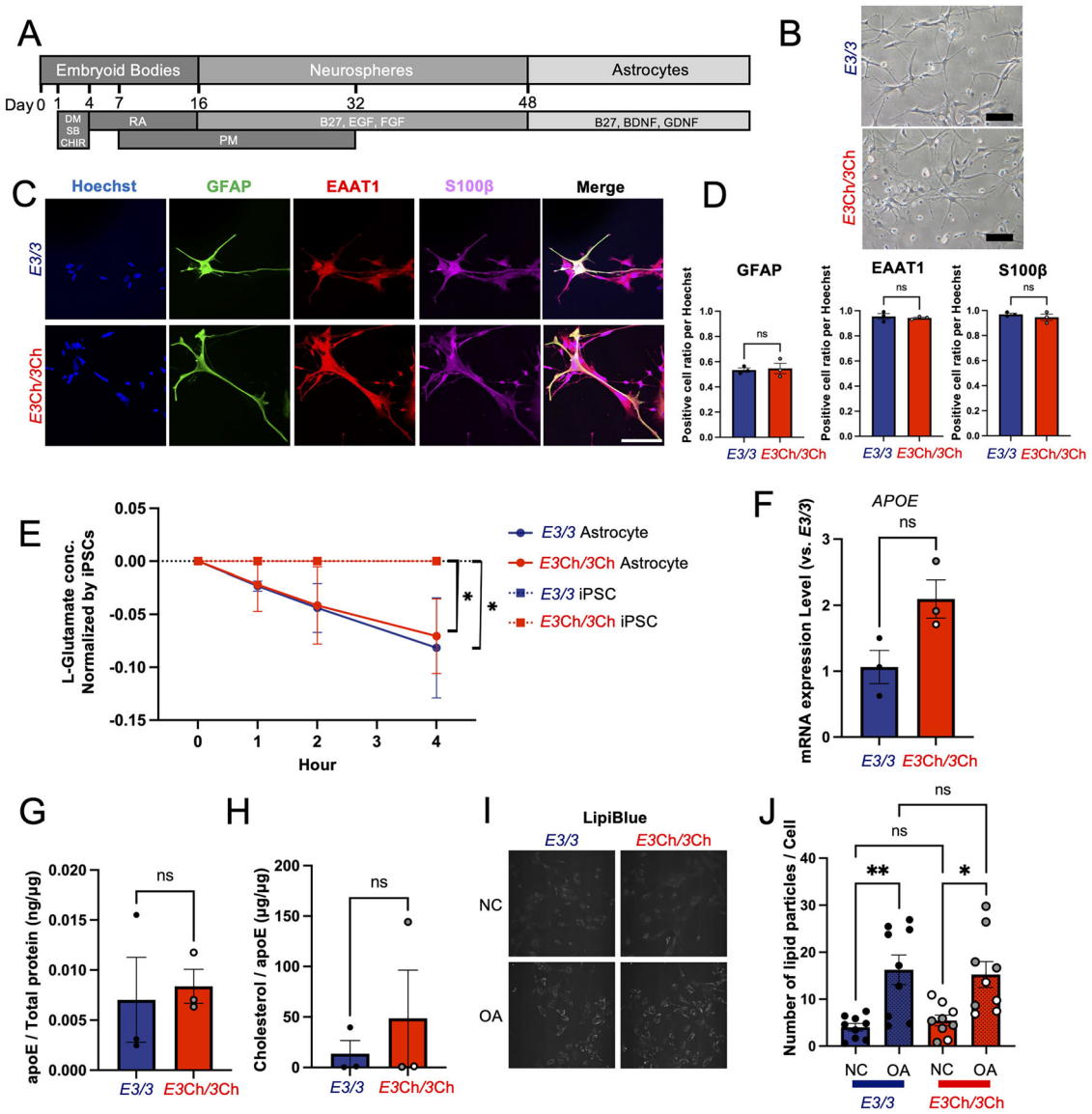


Figure 2. Differentiation from iPSCs into astrocytes carrying *APOE3/3* or *APOE3*Ch/3Ch and expression of apoE and lipid in *APOE3/3* and *APOE3*Ch/3Ch astrocytes. ***A***, Schematic protocol for differentiation into astrocytes. Each capital word represents as follows: DM, dorsomorphin; SB, SB431542; CHIR, CHIR99021; RA, retinoic acid; PM, purmorphamine; EGF, epidermal growth factor; FGF, fibroblast growth factor; BDNF, brain-derived neurotrophic factor; GDNF, glial cell line-derived neurotrophic factor. ***B***, Representative images of iPSC-derived astrocytes carrying *APOE3/3* or *APOE3*Ch/3Ch (clone #1) with bright fields. Scale bar: 100 μm. ***C***, Representative immunocytochemical images of iPSC-derived astrocytes carrying *APOE3/3* or *APOE3*Ch/3Ch (clone #1) stained with antibodies specific for GFAP, EAAT1, and S100β are shown. Scale bar: 100 μm. ***D***, Quantification of each astrocytic marker-positive cell per total cells from the immunocytochemical images of *APOE3/3* or *APOE3*Ch/3Ch (clone #1) indicated in ***C***. Data represent the mean ± SE. n.s., not significant, by unpaired *t* test. ***E***, L-glutamate concentration in culture media of iPSC-derived astrocytes or iPSCs carrying *APOE3/3* or *APOE3*Ch/3Ch (clone #1) are measured at 0, 1, 2, and 4 h after adding L-glutamate ($n = 3$ of independent culture batches). Data represent the mean ± S.E. $^*p < 0.05$ by two-way ANOVA versus respective iPSCs. ***F***, *APOE* gene expressed in *APOE3/3* and *APOE3*Ch/3Ch (labeled in gray for clone #1 or white for clone #7) astrocytes are analyzed by qRT-PCR ($n = 3$ of independent culture batches). Data represent the mean ± SE. n.s., not significant, by unpaired *t* test. ***G***, Secreted apoE measured by ELISA in conditioned media for *APOE3/3* and *APOE3*Ch/3Ch (labeled in gray for clone #1 or white for clone #7) astrocytes. Data represent the mean ± SE. n.s., not significant, by unpaired *t* test. ***H***, Cholesterol concentration of lipid fraction of IP using anti-apoE antibody against *APOE3/3* and *APOE3*Ch/3Ch (labeled in gray for clone #1 or white for clone #7) astrocytes. Data represent the mean ± SE. n.s., not significant, by unpaired *t* test. ***I***, Representative images of stained lipid particles in astrocytes with or without OA. Negative control. ***J***, Quantification of the number of lipid particles in *APOE3/3* and *APOE3*Ch/3Ch (labeled in gray for clone #1 or white for clone #7) astrocytes in ***H*** ($n = 9$ of independent batches). Data represent the mean ± SE. n.s., not significant, $^*p < 0.05$, and $^{**}p < 0.01$ by Tukey's test.

Immunocytochemistry of these cultures enabled us to find that GFAP-positive astrocytes formed different morphologies depending on the genotype, the cytokine treatment, and coculture with the iPSC-derived healthy control neurons (Fig. 3*D*). To quantify the morphological changes between *APOE3/3* and *APOE3*Ch/3Ch astrocytes, we performed Sholl analysis against GFAP-positive astrocytes (Fig. 3*E*). Although the *APOE3/3* and *APOE3Ch/3Ch* astrocytes exhibited similar morphologies, the cytokine treatment increased the morphological complexity only in the *APOE3/3* astrocytes but not in the *APOE3Ch/3Ch* astrocytes [Fig. 3*F*, *APOE3/3*; $F = 4.407$, degrees of freedom (df) 60/1,708, $p < 0.0001$, *APOE3Ch/3Ch*; $F = 0.8590$, df 60/1,708, $p = 0.7706$]. Taken together with the results of qRT-PCR in Figure 3*A–C*, these results indicate that the microglial cytokine induces a reactive state with a more complex morphology only in *APOE3/3* astrocytes. Both *APOE3/3* and *APOE3*Ch/3Ch astrocytes cocultured with the healthy control iPSC-derived neurons formed more complex morphologies than astrocyte culture

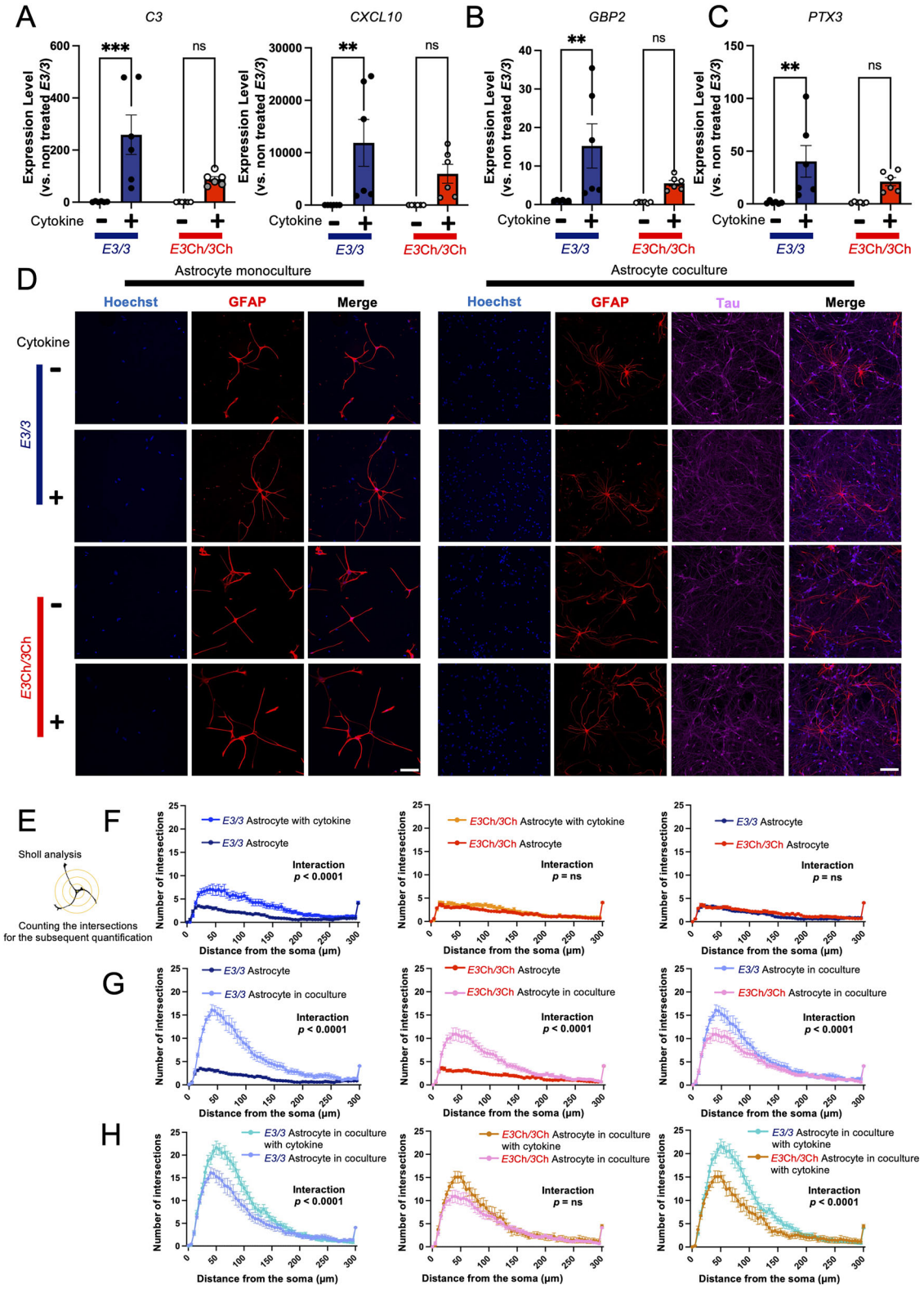


Figure 3. Alterations in gene expression and morphologies of *APOE3/3* and *APOE3*Ch/*3*Ch astrocytes. ***A–C***, Pan-reactive astrocyte marker genes (***A***), inflammatory-specific reactive astrocyte marker gene (***B***), and neurotrophic-specific reactive astrocyte maker gene (***C***) of *APOE3/3* and *APOE3*Ch/*3*Ch (labeled in gray for clone #1 or white for clone #7) astrocytes with cytokine treatment (TNFα, IL-1α, and C1q) are analyzed by qRT-PCR ($n = 6$ of independent culture batches). Data represent the mean ± SE. n.s., not significant, $*p < 0.05$ and $***p < 0.001$ by two-way ANOVA. ***D***, Representative immunocytochemical images of *APOE3/3* and *APOE3*Ch/*3*Ch astrocytic culture alone and astrocytes in coculture between human healthy control neurons treated with or without cytokines stained with antibodies specific for GFAP and tau. Scale bar: 100 μm. ***E***, The diagram of Sholl analysis for counting the intersections. ***F–H***, The results of Sholl analysis to GFAP-positive *APOE3/3* and *APOE3*Ch/*3*Ch (clone #1 and clone #7) astrocytes ($n = 15$ astrocytes in $n = 3$ independent culture batches). Data represent the mean ± SE. Interaction: n.s., not significant, and $p < 0.0001$ by two-way ANOVA.

alone, and in the coculture, *APOE3/3* astrocytes formed significantly more complex morphology than *APOE3*Ch/*3*Ch (Fig. 3*G*, $F = 1.992$, df 60/1,708, $p < 0.0001$). These results implied that coculture with human neurons increases their complexity not only in *APOE3/3* astrocytes but also in *APOE3*Ch/*3*Ch astrocytes, despite being much less complex in *APOE3*Ch/*3*Ch astrocytes. The cytokine treatment also enhanced the morphological complexity only in the cocultured *APOE3/3* astrocytes but not in the cocultured *APOE3*Ch/*3*Ch astrocytes (Fig. 3*H*, *APOE3/3*; $F = 3.323$, df 60/1,708, $p < 0.0001$, *APOE3Ch/3Ch*; $F = 1.155$, df 60/1,708, $p = 0.1973$). These results indicate that the cytokines induced *APOE3/3* astrocytes into a reactive state even in the coculture.

Taken together, the microglial cytokines can induce astrocytes into reactive astrocytes and enhance astrocytic morphological complexity. Coculture with healthy control neurons also enhances astrocytic morphological complexity. Together, *APOE3*Ch/*3*Ch astrocytes elicit minimal morphological change by the cytokines, implying that *APOE3*Ch/*3*Ch astrocytes have a resistance to convert into reactive astrocytes.

Tau propagation among *PSEN1* mutant neurons (H163R) via LRP1

The original report revealed that the case with *PSEN1* mutation (E280A) and *APOE* Christchurch variant had an atypical tau accumulation by means of tau PET imaging using flortaucipir and immunohistochemistry of the postmortem brain using anti-phosphorylated tau antibody AT8. These results suggest that *APOE* Christchurch suppressed tau propagation through neuronal circuit (Arboleda-Velasquez et al., 2019; Sepulveda-Falla et al., 2022).

To validate the hypothesis, we first established a tau propagation model, in which a vector consisting of a tandem sequence of NLS fused to dTomato and FLAG tag fused to human tau RD (residues 244th–372nd) was used. This vector can produce both NLS-dTomato and FLAG-tau RD by cleavage at P2A sequence. In the vector-transfected neurons, dTomato expresses at nuclear and FLAG-tau expresses where tau is distributed in the intracellular regions. Conversely, untransfected neurons principally show negligible protein expression by themselves. When FLAG-tau propagated from the vector-transfected neurons to the untransfected neurons, the propagated neurons express only FLAG-tau in the intracellular regions but not dTomato (Fig. 4*A*).

To enhance the tau secretion from neurons, we treated a high concentration of KCl in the vector-transfected iPSC-derived neurons carrying *PSEN1* mutation (H163R) and the healthy control iPSC-derived neurons for 1 h. ELISA for human total tau in the conditioned media of the neurons revealed that the depolarization stimulation significantly enhanced the tau secretion by about twofold only in the *PSEN1* mutant neurons (H163R), but not in the healthy control neurons (Fig. 4*B*).

A recent study reported that tau propagates via low-density lipoprotein receptor-related protein 1 (LRP1; Rauch et al., 2020). We first evaluated LRP1 expression in the iPSC-derived neurons. Immunocytochemistry of the *PSEN1* mutant neurons (H163R) revealed that the neurons expressed LRP1 (Fig. 4*C*). Further immunocytochemistry of the vector-transfected cultures showed that in addition to neurons expressing both dTomato and FLAG, some neurons expressed only FLAG, upon depolarization in the absence of RAP, which mimics the ligand-binding domain of LRP1 (Fig. 4*D*). These results demonstrated that FLAG-tau can propagate from the vector-transfected neurons to nearby neurons, which is significantly inhibited with RAP. dTomato- and FLAG double-positive neurons and only FLAG-positive neurons were counted, and the ratio of only FLAG-positive neurons per double-positive neurons was calculated as a tau propagation ratio. For the *PSEN1* mutant neuronal culture without RAP treatment, the KCl treatment significantly enhanced the tau propagation ratio compared with no treatment. RAP treatment significantly reduced the ratio among the KCl-treated *PSEN1* mutant neurons (Fig. 4*E*). Together, these results demonstrated that the secreted tau is incorporated via LRP1.

Establishment of coculture system between neurons and astrocytes as a tau propagation evaluation

To validate the tau propagation assay in coculture between neurons and astrocytes, we first examined neuronal viability and tau localization. From the immunocytochemistry and calcein staining images of *PSEN1* mutant neurons (H163R) alone and neurons cocultured with *APOE3/3* or *APOE3Ch/3Ch* astrocytes, the ratio of calcein-positive cells per total neurons did not change between the groups (Fig. 5*A*,*B*). Next, the expression level and subcellular localization of endogenous tau were evaluated by tau signal intensity in the whole neuronal area and soma, respectively. No difference in tau expression levels was observed in the whole neuronal area and soma (Fig. 5*A*,*C*). The secretion of different tau species was measured by ELISA between the neuronal culture alone and the cocultures. The coculture with astrocytes did not change the secretion levels of the total tau, phosphorylated tau at threonine 181, and phosphorylated tau at threonine 231 (Fig. 5*D*–*F*). These results suggest that astrocytic coculture does not affect neuronal survival, tau localization, and tau secretion.

Suppressive effects of *APOE3*Ch/*3*Ch astrocytes against tau propagation among *PSEN1* mutant neurons

To evaluate whether *APOE3*Ch/*3*Ch astrocytes have an ability to exert suppressive effects against tau propagation, we transfected the tau vector into the *PSEN1* mutant neurons (H163R or A246E) and healthy control neurons (1 or 2), followed by seeding astrocytes to the neurons. Immunocytochemistry revealed that some neurons expressed both dTomato and FLAG, and another set of neurons expressed only FLAG, mostly in KCl-treated cultures (Fig. 6*A*). First, the efficiencies of vector transfection into iPSC-derived neurons were calculated with the numbers of dTomato-positive neurons divided by Hoechst-positive neurons. The transfection efficiencies in *PSEN1* mutant neurons (H163R or A246E) and healthy control neurons (1 or 2) were comparable under KCl treatment and/or coculture (Fig. 6*B*). The coculture procedure did not seem to affect the distance between neurons, as the number of neurons was nearly confluent (Fig. 6*A*).

Next, we examined tau propagation in these neural cultures. For *PSEN1* mutant neuronal culture alone (H163R or A246E), the KCl treatment significantly increased the tau propagation ratio compared with no treatment (Fig. 6*C*). The KCl treatment also increased the tau propagation ratio among the *PSEN1* mutant neurons cocultured with *APOE3/3* astrocytes, although it was significantly lower than in neuronal culture alone. Interestingly, the *PSEN1* mutant neurons cocultured with *APOE3*Ch/*3*Ch astrocytes had the lowest tau propagation ratio among KCl-treated groups. Furthermore, in the *APOE3*Ch/*3*Ch astrocyte-cocultured neurons, KCl treatment had a minimal

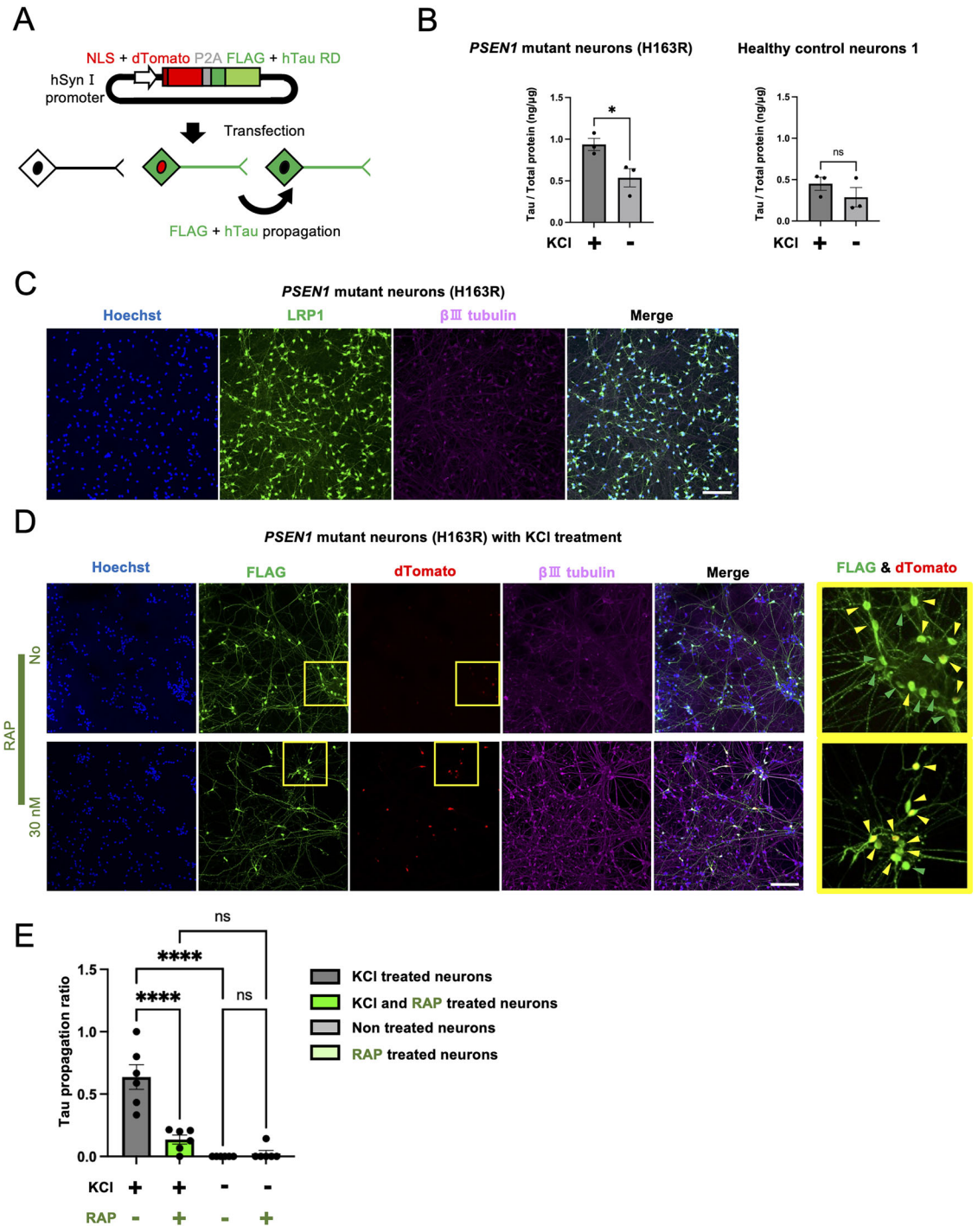


Figure 4. Tau propagation among *PSEN1* mutant (H163R) neurons via LRP1. ***A***, Schematic diagram for the establishment of the tau propagation model. The vector contains human *synapsin I* promoter and sequences of NLS fused to dTomato, P2A, and FLAG tag fused to human tau RD. This vector can produce NLS-dTomato and FLAG-tau RD in transduced neurons. The vector-transfected neurons express both exogenous proteins, but untransfected neurons do not express them. When FLAG-tau propagates to vector-untransfected neurons, the nearby neurons only exhibit FLAG-tau protein. ***B***, Secreted total tau in the conditioned media of *PSEN1* mutant neurons and healthy control neurons 1 with or without KCl treatment are measured by ELISA ($n = 3$ of independent culture batches). Data represent the mean ± SE. n.s., not significant, $^{*}p < 0.05$ by unpaired *t* test. ***C***, Representative immunocytochemical images of *PSEN1* mutant neurons (H163R) stained with antibodies specific for LRP1 and βIII tubulin. Scale bar: 100 μm. ***D***, Representative immunocytochemical images of the KCl treated vector-transfected *PSEN1* mutant neurons (H163R) alone with RAP treatment stained with antibodies specific for FLAG and βIII tubulin along with the intrinsic fluorescence of dTomato. Scale bar: 100 μm. Yellow arrowheads indicate the vector-transfected neurons and green arrowheads indicate tau-propagated neurons in a larger image of yellow frame regions merged with FLAG and dTomato. ***E***, Quantification of tau propagation ratio among *PSEN1* mutant neurons (H163R) with or without KCl and RAP treatments ($n = 3$ of independent batches). Data represent the mean ± SE. n.s., not significant, and $^{****}p < 0.0001$ by Tukey's test.

effect on tau propagation compared with the no treated group. The tau propagation ratio among the healthy control neurons (1 or 2) showed no change with or without astrocyte culture and KCl treatment (Fig. 6C). These data suggest that *APOE3/3* astrocytes can fundamentally suppress the tau propagation among *PSEN1* mutant neurons (H163R or A246E), and

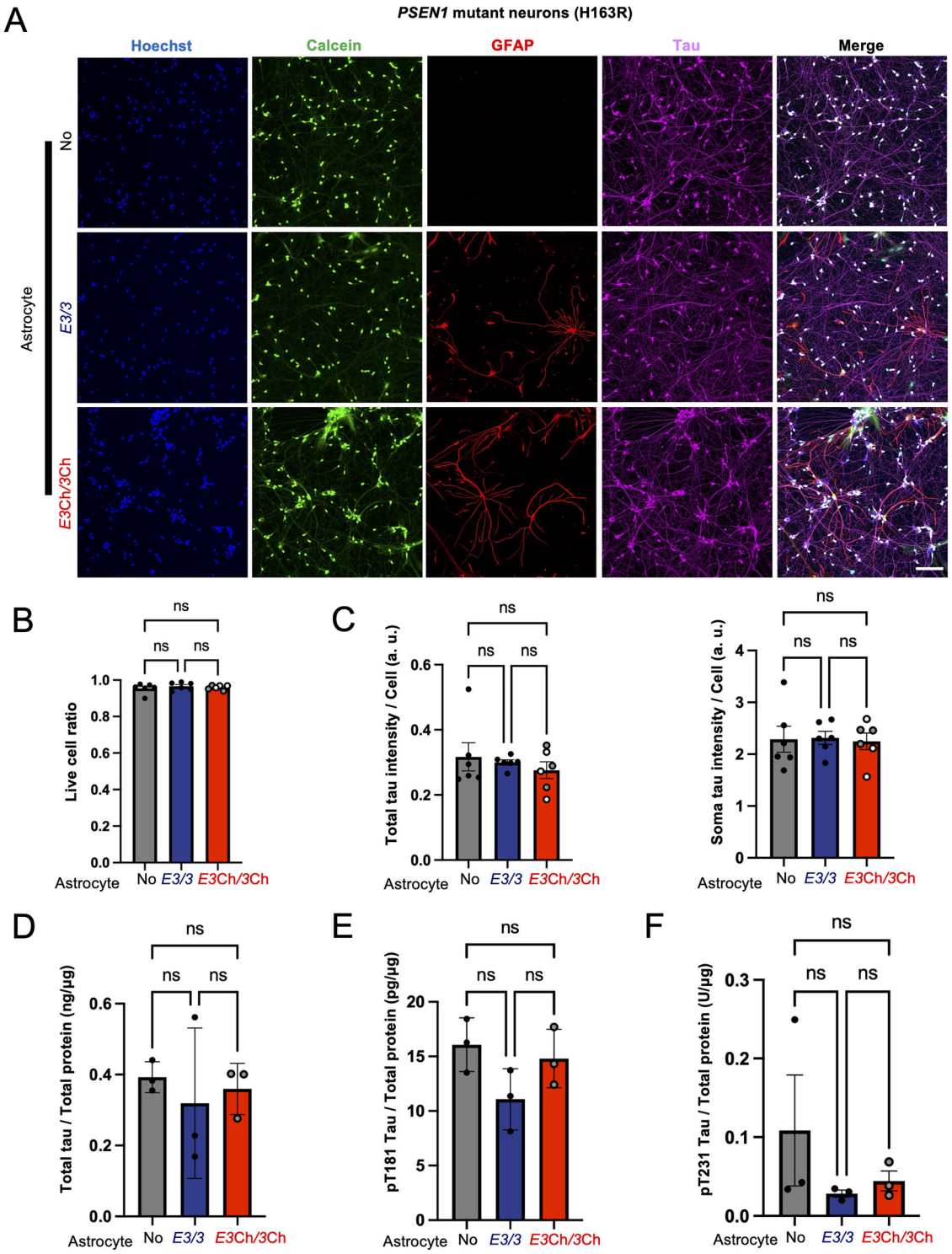


Figure 5. Validation of *PSEN1* mutant neurons (H163R) alone and coculture with *APOE3/3* or *APOE3*Ch/3Ch astrocytes. ***A***, Representative immunocytochemical images of *PSEN1* mutant neurons (H163R) alone and coculture with *APOE3/3* or *APOE3*Ch/3Ch astrocytes stained with calcein-AM and antibodies specific for GFAP and tau. Scale bar: 100 μm. ***B***, Quantification of survival cells using the numbers of calcein-positive cells and Hoechst-positive cells in the *PSEN1* mutant neurons (H163R) alone and coculture with *APOE3/3* or *APOE3*Ch/3Ch (labeled in gray for clone #1 or white for clone #7) astrocytes. Data represent the mean ± SE. n.s., not significant, by Tukey's test. ***C***, Quantification of tau intensities in the whole cells (left) and at the soma (right) in the *PSEN1* mutant neurons (H163R) alone and coculture with *APOE3/3* or *APOE3*Ch/3Ch (labeled in gray for clone #1 or white for clone #7) astrocytes. Data represent the mean ± SE. n.s., not significant, by Tukey's test. ***D–F***, Secreted total tau (***D***), phosphorylated T181 tau (***E***), or phosphorylated T231 tau (***F***) in the conditioned medium of *PSEN1* mutant neurons (H163R) alone and coculture with *APOE3/3* or *APOE3*Ch/3Ch (labeled in gray for clone #1) astrocytes treated with KCl are measured by ELISA ($n = 3$ of independent culture batches). Data represent the mean ± SE. n.s., not significant, by Tukey's test.

*APOE3*Ch/*3*Ch astrocytes have a further suppressive function against the tau propagation in neurons with extinguishing the effect of depolarization stimulation by KCl.

Discussion

The *APOE* Ch variant was first reported as a risk factor for hyperlipoproteinemia type III in 1987 from Christchurch, New

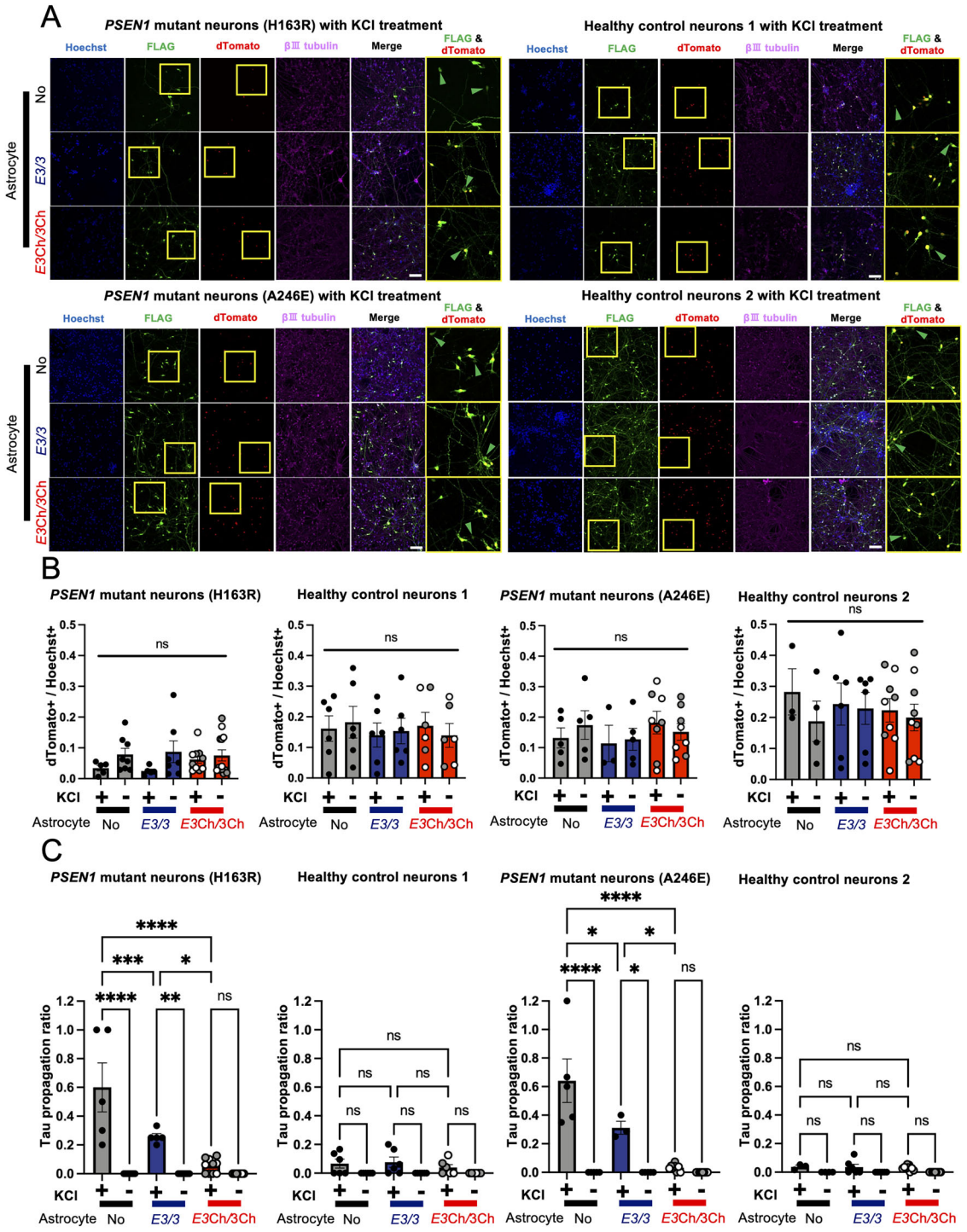


Figure 6. Suppressive function against tau propagation among *PSEN1* mutant neurons by *APOE*3Ch/3Ch astrocytes. ***A***, Representative immunocytochemical images of the vector-transfected *PSEN1* mutant neurons (H163R), healthy control neurons 1, *PSEN1* mutant neurons (A246E), and healthy control neurons 2 in the conditions of neuronal culture alone and coculture with *APOE3/3* or *APOE3*Ch/*3*Ch (clone #1 or clone #7) astrocytes treated with KCl stained with antibodies specific for FLAG and βIII tubulin along with the intrinsic fluorescence of dTomato. Scale bar: 100 µm. Green arrowheads indicate tau-propagated neurons in a larger image of yellow frame regions merged with FLAG and dTomato. ***B***, The result of calculations of vector transfection ratio to the *PSEN1* mutant neurons (H163R or A246E) and the healthy control neurons (1 or 2) alone or *APOE3/3* or *APOE3*Ch/*3*Ch (labeled in gray for clone #1 or white for clone #7) astrocyte coculture with or without KCl treatment ($n = 3$–6 of independent batches). Data represent the mean ± SE. n.s., not significant, by Tukey's test. ***C***, Quantification of tau propagation ratio among the *PSEN1* mutant neurons (H163R or A246E) and the healthy control neurons (1 and 2) alone or *APOE3/3* or *APOE3*Ch/*3*Ch (labeled in gray for clone #1 or white for clone #7) astrocyte coculture with or without KCl treatment ($n = 3$–6 of independent batches). Data represent the mean ± SE. n.s., not significant, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$ and $****p < 0.0001$ by Tukey's test.

Zealand (Wardell et al., 1987). Since then, a few kindreds carrying the *APOE* Ch variant have been reported as a risk factor for hyperlipoproteinemia, but not as a relevant factor for other diseases (Pocovi et al., 1996; Feussner et al., 1998; Rolleri et al., 2003). The recent case report is the first to explain an association between the *APOE* Ch variant and AD pathogenesis. This case suggested the protective effects of the *APOE* Ch variant against familial AD onset with prevention of tau pathology from

spreading throughout the brain (Arboleda-Velasquez et al., 2019; Sepulveda-Falla et al., 2022). However, it was still unclear whether the *APOE* Ch variant suppressed the exaggeration of tau pathology. Human iPSC-derived models, specifically astrocytes and neurons, have been used to analyze AD-related pathologies associated with *APOE4*, *APOE3*, and *APOE2* (Lin et al., 2018; Sienski et al., 2021; de Leeuw et al., 2022; Tcw et al., 2022; Watanabe et al., 2023).

Although human iPSC-derived modeling is a prominent tool for the analysis of apoE function and AD pathology, only two studies have examined the *APOE* Ch variant in iPSC-derived models. These studies demonstrated the ability of the *APOE3*Ch/*3*Ch to mitigate phosphorylated tau accumulation in the context of the *PSEN1* Colombian mutation (Mazzarino et al., 2023), as well as the attenuated tau uptake by neurons carrying the *APOE4*Ch/*4*Ch genotype (Nelson et al., 2023). Furthermore, recent mouse studies demonstrated that *APOE* Ch suppresses *APOE4*-related and Aβ-induced tau pathology (Nelson et al., 2023; Chen et al., 2024). However, no report has focused on detailed astrocytic functions of the *APOE* Ch variant. In this study, to elucidate the protective functions of the *APOE* Ch variant against AD development, we generated isogenic *APOE3*Ch/*3*Ch iPSCs from parental *APOE3/3* healthy control iPSCs using a CRISPR/Cas9 genome editing system and differentiated them into astrocytes. We validated the functions of *APOE3*Ch/*3*Ch iPSC-derived astrocytes and first demonstrated *APOE3*Ch/*3*Ch astrocytes' ability to repress interneuronal propagation of tau. These results provide the first experimental evidence to uncover the underlying mechanism behind the pathological findings observed in the first report (Arboleda-Velasquez et al., 2019).

Changes in lipid metabolism such as cholesterol accumulation in iPSC-derived astrocytes and apoE oligomerization and lipidation are related to *APOE4* or *APOE3* Jac variant, which could be crucial causes of AD pathology (Lin et al., 2018; C. C. Liu et al., 2021; Sienski et al., 2021; Tcw et al., 2022). However, our iPSC-derived astrocyte model revealed no difference in lipid accumulation and cholesterol secretion between *APOE3/3* and *APOE3*Ch/*3*Ch astrocytes with or without OA. Our astrocytes were differentiated and maintained with serum-free media, but astrocytes of the other groups reporting astrocytic AD lipid phenotypes were maintained with media including serum. Unidentified components in serum may play important roles in astrocytic lipid metabolism as AD pathologies (Lin et al., 2018; Tcw et al., 2022). Our serum-free condition may cause no difference in lipid accumulation and secretion between *APOE3/3* and *APOE3*Ch/*3*Ch astrocytes. Although changes in lipid metabolism were not observed, our *APOE3*Ch/*3*Ch astrocyte model demonstrated suppression against tau propagation. This study suggests that the protective effect of the *APOE* Ch variant is an independent phenomenon from serum in media and changes of lipid metabolism.

*APOE3*Ch/*3*Ch astrocytes formed less complex morphology with the microglial cytokine treatment and in a coculture with iPSC-derived neurons (Fig. 3). Some reports suggests that reactive astrocytes formed a more complex and hypertrophic morphology than nonreactive astrocytes (Wilhelmsson et al., 2006; Chun et al., 2020). These indicated that the *APOE3*Ch/*3*Ch astrocytes had lower potential to convert into reactive astrocytes with the cytokine treatment and in the coculture system. A morphological change of *APOE3*Ch/*3*Ch astrocytes was consistent with the results of qRT-PCR for cytokine-treated astrocytes. Taken together, these results suggest that *APOE3*Ch/*3*Ch astrocytes tend to resist the transition to a reactive state upon inflammatory treatment, and it may be related to one of the protective aspects against AD pathology.

The *APOE* Ch variant could influence tau pathology as shown in the first case (Arboleda-Velasquez et al., 2019). In vivo tau pathology is strongly correlated with the ability of tau propagation across brain regions (de Calignon et al., 2012; L. Liu et al., 2012). Therefore, we first developed a cellular assay for tau propagation into nearby naive neurons using an exogenous reporter vector, in which neurons exhibiting a sole FLAG-positive signal are considered recipients of the propagated tau (Fig. 4*D*). Wegmann et al. (2019) have reported a tau propagation reporter using a similar construct: an adeno-associated virus consisting of a fluorescent protein and full-length human tau. This construct has been used to demonstrate tau propagation both in culture and in mice (Wegmann et al., 2019; Rauch et al., 2020; Caballero et al., 2021; Dujardin et al., 2022; Rodrigues et al., 2023). However, we observed tau propagation among human iPSC-derived neurons using a tau RD (aa 244–372) fused to the FLAG tag. Our results suggest that tau RD is sufficient to assess the ability of tau to propagate to nearby iPSC-derived neurons. The molecular mechanisms of tau propagation are still unknown (Takeda, 2019; Lamontagne-Kam et al., 2023), but neuronal incorporation of extracellular tau appears to be mediated by the RD region. A recent report shows that a high content of lysine residues in the RD region is crucial for binding to LRP1, a putative tau receptor for propagation (Rauch et al., 2020). Furthermore, tau propagation was significantly suppressed by an administration of antibody D against the tau epitope aa 235–250, which recognizes the tau RD region (Albert et al., 2019). Together with the previous reporter using full-length tau, these types of reporters may prove to be invaluable tools for performing a sensitive tau propagation assay in different contexts, including human neurons as investigated in this study.

In this study, we were unable to demonstrate apparent tau propagation in iPSC-derived neurons without depolarization stimulation. This claim is supported by the report that neuronal activity stimulates tau propagation in vitro and enhances tau pathology in vivo (Wu et al., 2016). Indeed, we found that depolarization stimulation increased secreted tau levels as shown in previous reports (Pooler et al., 2013; Yamada et al., 2014). However, we observed that tau secretion was enhanced by depolarization stimulation in the *PSEN1* mutant neurons but not in the healthy control neurons (Fig. 4*B*). This suggests that *PSEN1* mutant neurons exhibit precocious differentiation, acquiring the property for high tau secretion by depolarization stimulation (Arber et al., 2021).

Intriguingly, we demonstrated the suppressive effects of astrocytes carrying the *APOE* Ch variant against tau propagation in human neurons. Astrocytes fundamentally have the ability to suppress exogenous tau propagation among *PSEN1* mutant neurons (H163R or A246E), and *APOE3*Ch/*3*Ch astrocytes further suppressed tau propagation among *PSEN1* mutant neurons, which are not the same *PSEN1* (E280A) mutation as in the case report (Arboleda-Velasquez et al., 2019). Suppressive functions of *APOE3*Ch/*3*Ch astrocytes are likely to be independent of *PSEN1* mutation types in neurons, but dependent on the presence of the *PSEN1* mutation. Recent reports using mice and iPSC-derived models demonstrated that *APOE* Ch variant attenuated AD phenotypes in the context of distinct *APOE* and/or *PSEN1* genotypes (Mazzarino et al., 2023; Nelson et al., 2023; Chen et al., 2024). In this study, our findings are the first report to show the suppressive effects of astrocytic *APOE3*Ch variant on

tau propagation among *PSEN1* mutant neurons. However, it is still controversial whether other genotypes of the *APOE* gene, *E2*, *E3*, and *E4*, affect tau propagation (Williams et al., 2022; Davies et al., 2023). We observed prevention by *APOE3*Ch/*3*Ch astrocytes only in *PSEN1* mutant neurons, suggesting a unique function of *APOE* Ch variant against tau pathology. Further studies are needed to elucidate the underlying suppressive mechanism of *APOE* Ch variant and the effect of other *APOE* variants on tau propagation, possibly contributing to the development of therapies for AD onset and progression. Our iPSC-derived models are fundamental tools for clinical drug development.

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